MLTEST -Report

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Diagnosis of Colorectal Cancer by identification of microbiome species in group of patients

Main Objective

Build ML classification algorithm that can predict patient status from their relative species abundances, mainly Cancer or not Cancer

Data Preparation

-Microbiome species abundance data for 1981 species and 187 patients

-Meta data figuring notably the « diagnosis » case (87: Cancer, 100: not Cancer)

Main Step

Perform machine learning (ML) based analyses on the species abundances and propose a « signature » model of CRC pathology

ML Model Building and Evaluation In Rstudio

3 machine learning models (supervised models) were adapted and trained on the whole dataset to predict the status of each patient.

Models evaluation and selecting the best model.

Random Forest

- perform caret package (**train** function with 5-fold cross-validation).
- Calculating area under the rock curve by applying cross-validation.
- Extracting the plots and results of all important species.

Logistic Regression

- perform caret package (**train** function with 5-fold cross-validation).
- Calculating area under the rock curve by applying cross-validation.
- Extracting the plots and results of all important species.

Logistic Regression with lasso Penalty by 2 packages

Lasso model with <u>Glmnet package</u>

- Apply **cv.glmnet** function to find the best lambda (penalization parameter and level).
- plot the Lasso models with lambda values
- Evaluation of Lasso model with the optimal lambda

<u>Lasso Model with caret Package</u>

- Apply the regularization by applying **train** function using 5-fold cross-validation.
- Apply lambda values generated from **cv.glmnet function.**
- Generate a plot of the Lasso model
- Evaluate the Lasso model.

<u>Secondery Objective</u>

Network Visualization of Important Species

Representation of the different species involved in the signature of CRC in the form of a network using species distances based upon their role in the RF model

Create two network representations:

Simple Network

- Selecting the higher important species by selecting the higher score;
  from the Random Forest model select the first 40 markers for example.
- Generate an adjacency matrix based on a threshold of importance scores.
- Plot a network (nodes were colored based on importance scores)

Network based on Distances

- Calculate distances between higher important species based on the scores.
- Set a distance threshold (here I selected randomly) and generated an adjacency matrix.
- Plot a network with nodes representing species and edges representing connections based on distances.